Spatial model of Real Estate portfolio exposure using Cholesky decomposition -

$$Y(x) = ay(x) + bw(x) \int_0^T Z_w(sx)\, ds + Z_y(Tx),$$

where $Z_y(tx)$ is a partly time-changed Brownian sheet

$$Z_y(tx) = y \exp\left(y_1 x_1 + y_2 x_2\right) W_t^{v_y},$$

with the time changes

$$v_y^2(x_1) = \exp(2y_1 x_1), \quad v_y^3(x_2) = \exp(2y_2 x_2),$$

- deterministic component $ay(x)$, the integral of temperature deviations from the trend $\int_0^T Z_w(sx)\, ds$ weighed by a factor $bw(x)$, and a noise term $Z_y(Tx)$.

$$\mathrm{Cov}[Z_y(tx) Z_y(sy)] = 2 \exp\left(y_1(t-s) + y_2(x_2 - y_2)\right).$$

$x_f = (x_1, \ldots, x_d)$ are geographical positions. If the correlation matrix of $Z_y(tx_f)$ has the Cholesky decomposition

$$\mathrm{corr}(Z_y(tx_f), Z_Y(tx_k)) = \exp(y_1(x_{f1} - x_{k1}) + y_2(x_{f2} - x_{k2})),$$

then the vector $Z_f y(t)$ has the following representation

$$Z_f y(t) = Z_f y(0) + y W_t^y,$$

where W_t^y is a vector of independent standard Brownian motions. $w \in \mathbb{R}^{p+d}$ is the Cholesky decomposition of the spatial correlation matrix of $Z_w(tx)$ at locations $x_1, \ldots, x_p$. Diagonal matrices

$$b_w = \mathrm{diag}(b_w(x_f)), \quad h_w = \mathrm{diag}(w(x_h)), \quad f_w = \mathrm{diag}(w(x_f)),$$

RE thus follows a multivariate normal distribution with moments -

$$E[Y|F_t] = ay + bw \int_0^T Z_f w(s)\, ds + bw(1 - e^{w(T-t)}) Z_f w(t) + Z_f y(t),$$

and

$$\mathrm{Var}[Y|F_t] = \int_0^t (1 - e^{w(T-s)})^2 ds + bw f_w (h_w^T + f_w^T) f_w b_w + 2y(T-t) y^T y.$$